

Marketing Research and Data Analytics

Learning Outcomes

To understand the role of market research in business decisions through the systematic and objective collection of data.

To comprehend the applications of market research and understand the various studies that analyses the size of markets, to measure the satisfaction of customers with products, to guide new product development and to show people's use of and attitudes to products.



Definitions of Market Research

Kotler (1999) defines marketing research as _systematic problem analysis, model-building, and fact-finding for improved decision-making and control in the marketing of goods and services.

The American Marketing Association (AMA, 1961) defines it as the systematic gathering, recording, and analyzing of data relating to the marketing of goods and services.

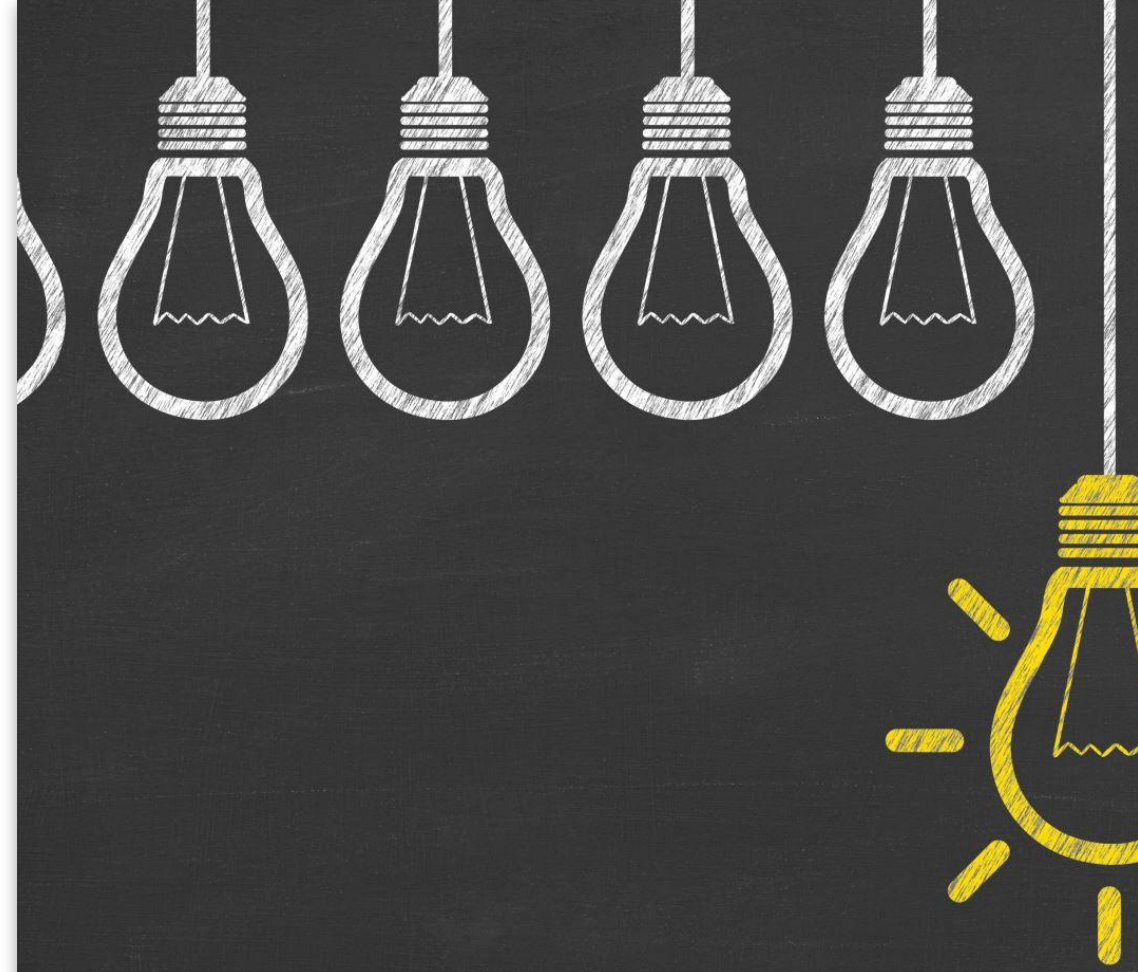
Marketing research (also called **consumer research**) is a form of business research.

- Marketing research is a systematic and objective study of problems in the marketing of goods and services. It applies to any area of marketing.
- Research is the only tool an organization has to keep in contact with its external operating environment.

Scope of marketing research

Marketing research can be used in

- Product Management – Competitive intelligence, pre-launch strategy for new products- test marketing and concept testing
- Sales analysis
- Corporate Research
- Advertising Research
- Syndicated Research





All of these forms of marketing research can be classified as either ***problem-identification research*** or as ***problem-solving research***.



A similar distinction exists between *exploratory research* and *conclusive research*.



Exploratory research provides insights into and comprehension of an issue or situation. It should draw definitive conclusions only with extreme caution.



Conclusive research concludes: the results of the study can be generalized to the whole population.

WHAT'S THE PROBLEM WITH A WATERMELON?

Consumer research at Syngenta and Seminis, both large seed companies, found that for single people, couples, and small families, watermelons are just too big. Even for larger families, there was an objection both to the way watermelons take up too much space in the refrigerator and the fact that they are too heavy to take on picnics. The result of this consumer research was the development of 'personal' or 'mini' watermelons. They weigh only three to five pounds and are seedless and sweeter than traditional melons. However, they cost two to three times per pound more than traditional melons.

Has this stopped consumers from purchasing? No – in fact when the mini-melons were introduced, demand exceeded supply. Consumers were willing to pay more for the benefits they desired. Yet no one would have known about these consumer desires without first conducting marketing research.





‘Identify and define marketing opportunities and problems’ using research to explore the external environment.



‘Generate, refine and evaluate marketing actions’ using research to determine whether the company is meeting consumer needs.

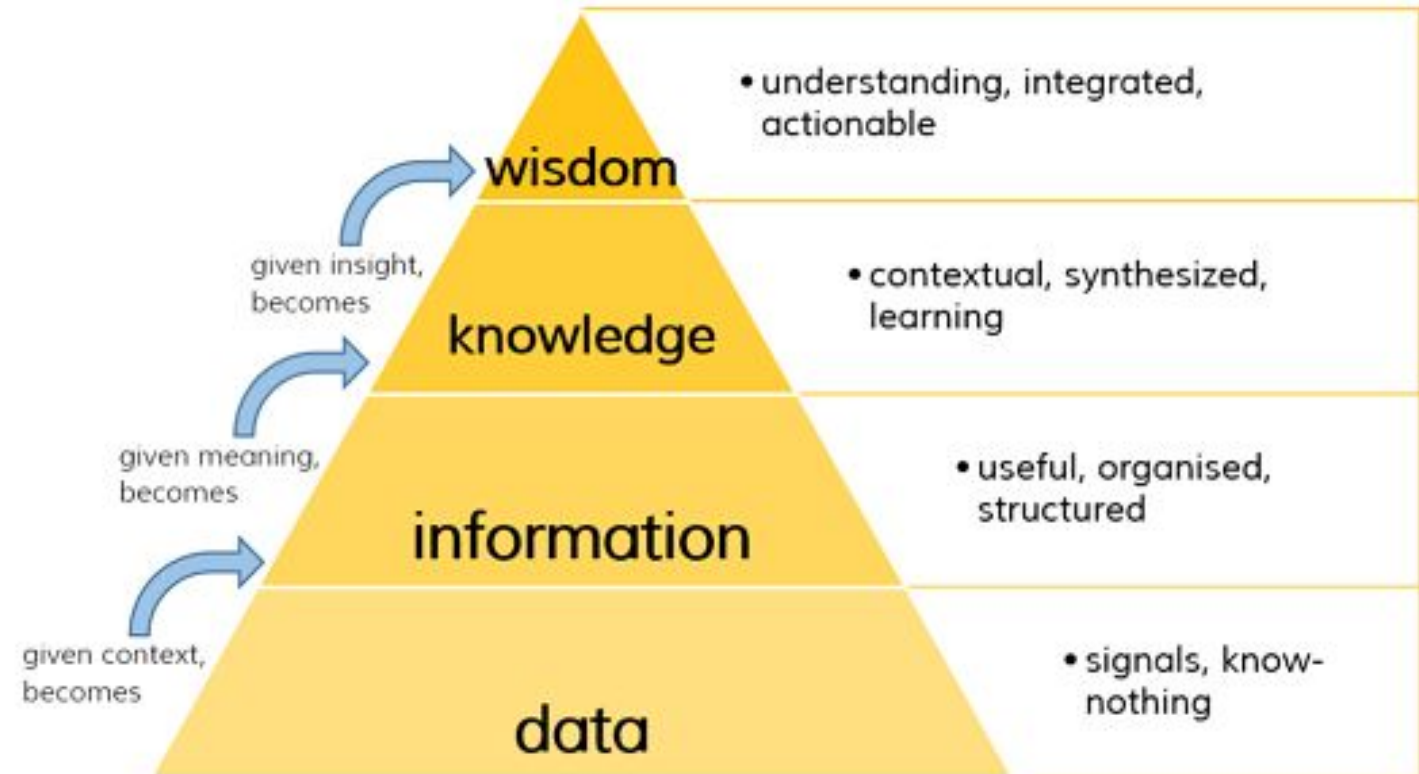
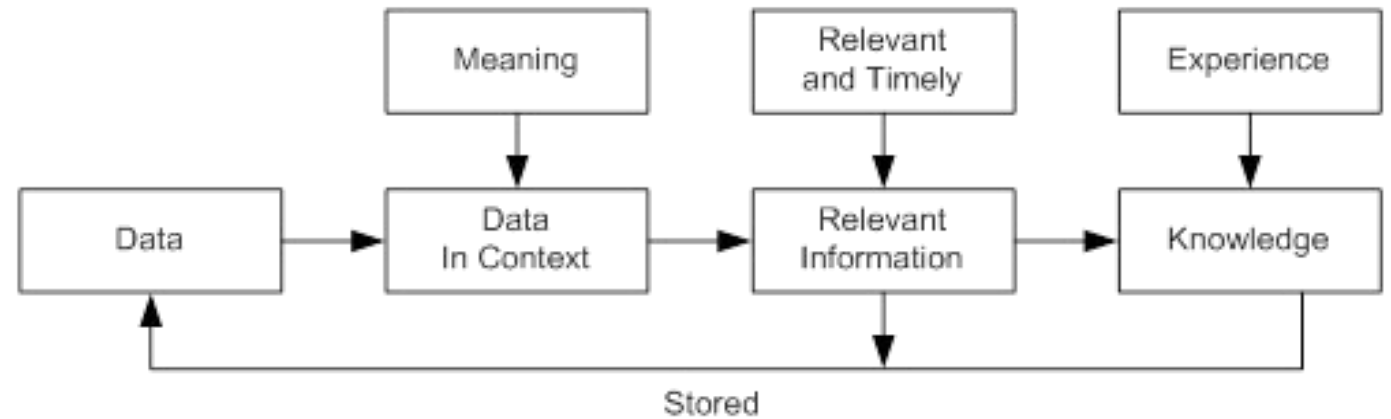


Monitor marketing performance’ using research to confirm whether the company is meeting the goals it has set.



‘Understanding marketing as a process’ using research to learn to market more effectively.

The relationship between data, information and knowledge



Marketing Research Methods

Methodologically, marketing research uses four types of research designs, namely:

Qualitative marketing research – This is generally used for exploratory purposes. The data collected is qualitative and focuses on people's opinions and attitudes towards a product or service. The respondents are generally few in number and the findings cannot be generalized to the whole population. No statistical methods are generally applied.

Ex: Focus groups, In-depth interviews, and Projective techniques

Quantitative marketing research – This is generally used to conclude a specific problem. It tests a specific hypothesis and uses random sampling techniques so as to infer from the sample to the population. It involves a large number of respondents and analysis is carried out using statistical techniques.

Ex: Surveys and Questionnaires



Observational techniques - The researcher observes social phenomena in their natural setting and draws conclusion from the same. The observations can occur **cross-sectionally** (observations made at one time) or **longitudinally** (observations occur over several time-periods) Ex: Product-use analysis and computer cookie tracing

Experimental techniques – Here, the researcher creates a quasi-artificial environment to try to control spurious factors, then manipulates at least one of the variables to get an answer to a research

Ex: Test marketing and Purchase laboratories



The meaning of research is to investigate thoroughly. It can be divided into basic and applied research.

Basic research is pure research which is more educative. Applied research is carried out to seek alternate solutions for a problem.

Applied research can further be classified as problem-solving research and problem-oriented research depending on the research problem at hand.

- Marketing research is a systematic and objective study of problems pertaining to the marketing of goods and services. It applies to any area of marketing. For ex. Product management, sales, advertising research, etc.
- Marketing Research can be Primary Market Research or Secondary Market research depending on the data source used. It can be qualitative or quantitative research depending upon the nature of the research.
- Marketing Research is different from Market Research wherein the former is oriented towards solving marketing problems and the latter is market related.

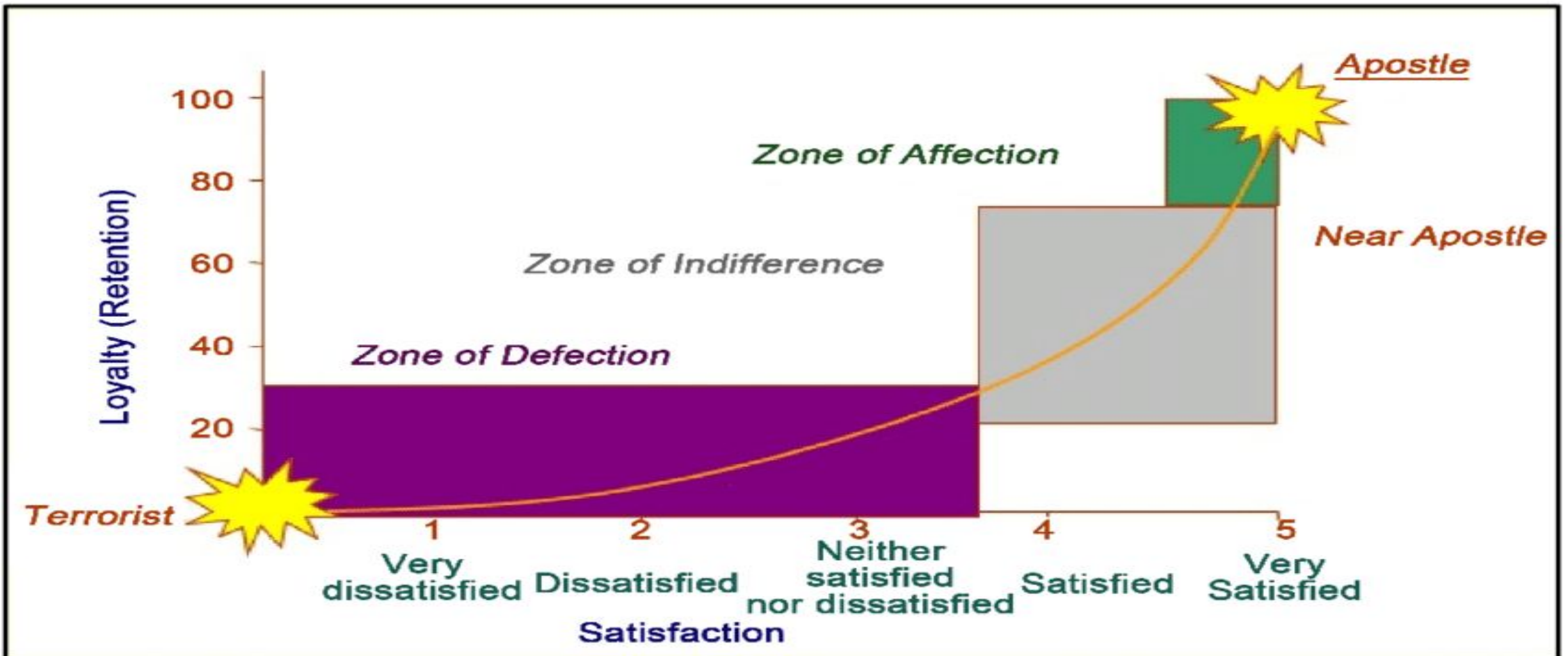
Difference between Market Research and Marketing Research

Market Research: Market Research involves researching a specific industry or market. Ex: Researching the automobile industry to discover the number of competitors and their market share.

Marketing Research: Marketing Research analyses a given marketing opportunity or problem, defines the research and data collection methods required to deal with the problem or take advantage of the opportunity, through to the implementation of the project. It is a more systematic method which aims to discover the root cause for a specific problem within an organisation and put forward solutions to that problem.

Ex: Research carried out to analyze and find solution for increasing turnover in an organisation.

The link between loyalty and customer satisfaction



Sources of Market Data

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graph TD; A[Sources of Market Data] -- INTERNAL --> B[Internal Sources (within the org.)]; A -- EXTERNAL --> C[Primary Sources]; A -- EXTERNAL --> D[Secondary Sources]; B --> B1[Accounting Information]; B --> B2[Sales Reports]; B --> B3[Expenditure data of various types]; B --> B4[Statistics of various kinds]; C --> C1[Salesmen]; C --> C2[Dealers]; C --> C3[Consumers]; D --> D1[Periodicals and Newspapers]; D --> D2[Govt. Publications and Reports]; D --> D3[Published Market Surveys]; D --> D4[Data released by International agencies];
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INTERNAL

EXTERNAL

Internal Sources (within the org.)

Accounting
Information

Sales Reports

Expenditure data of
various types

Statistics of various
kinds

Primary Sources

Salesmen

Dealers

Consumers

Secondary Sources

Periodicals and
Newspapers

Govt. Publications and
Reports

Published Market
Surveys

Data released by
International agencies

The Marketing Research process

1. Identify the problem

- Problem to solve
- Project objectives
- Research questions

2. Develop the research plan

- Information needed
- Research & analysis methods
- Responsible parties

3. Conduct research

- Secondary data review
- Primary data collection
- Suitable methods & techniques

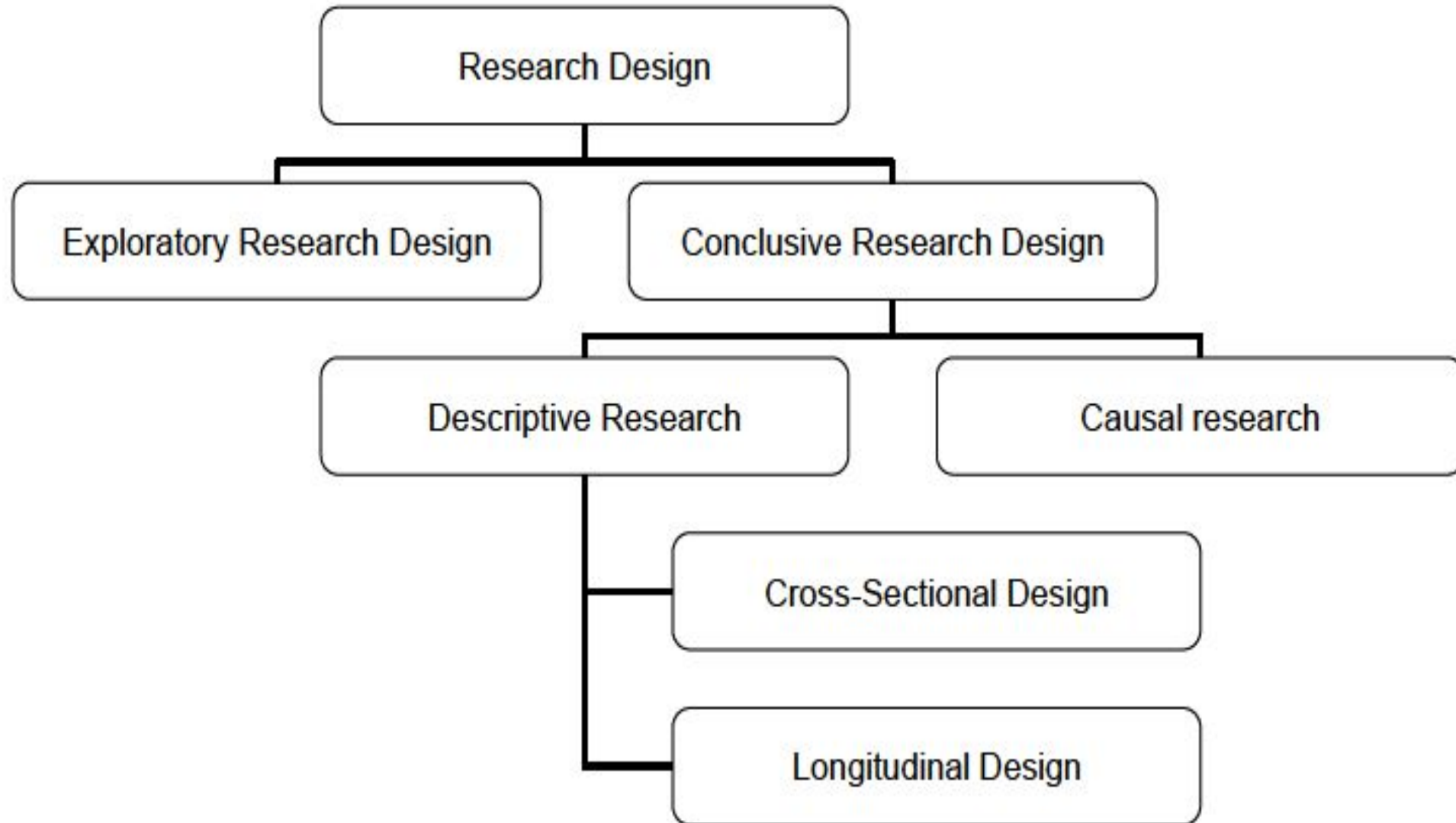
4. Analyze and report findings

- Data formatting & analysis
- Interpretation of results
- Report & recommendations

5. Take action

- Thought and planning
- Evaluation of options
- Course adjustment & execution

Classification of Research design



Exploratory research refers to the investigation of a research problem that is not clearly defined or understood.

Conclusive research, as the name suggests, is a type of research that is used to yield information that helps reach conclusions or make decisions.

Descriptive Research - The researcher is interested in describing a case, situation, or phenomenon, they are acting under a descriptive research design.

Causal research, also known as explanatory research or causal-comparative research, identifies the extent and nature of cause-and-effect relationships between two or more variables.

A cross-sectional study is a type of research design in which you collect data from many different individuals at a single point in time.

In a **longitudinal study**, you repeatedly collect data from the same sample over an extended period.

Questionnaire

TABLE 4.1

Types of Questions

Name	Description	Example
A. Closed-End Questions		
Dichotomous	A question with two possible answers	In arranging this trip, did you personally phone American? Yes _____ No _____
Multiple choice	A question with three or more answers	With whom are you traveling on this flight? ☐ No one ☐ Children only ☐ Spouse ☐ Business associates/friends/relatives ☐ Spouse and children ☐ An organized tour group
Likert scale	A statement with which the respondent shows the amount of agreement/disagreement	Small airlines generally give better service than large ones. Strongly disagree Disagree Neither agree nor disagree Agree Strongly agree 1 _____ 2 _____ 3 _____ 4 _____ 5 _____
Semantic differential	A scale connecting two bipolar words. The respondent selects the point that represents his or her opinion.	I find American Airlines ... Large _____ Small Experienced _____ Inexperienced Modern _____ Old-fashioned
Importance scale	A scale that rates the importance of some attribute	Airline in-flight service to me is Extremely important Very important Somewhat important Not very important Not at all important 1 _____ 2 _____ 3 _____ 4 _____ 5 _____
Rating scale	A scale that rates some attribute from "poor" to "excellent"	American in-flight service is Excellent Very Good Good Fair Poor 1 _____ 2 _____ 3 _____ 4 _____ 5 _____
Intention-to-buy scale	A scale that describes the respondent's intention to buy	If ultra high-speed Wi-Fi service were available on a long flight, I would Definitely buy Probably buy Not sure Probably not buy Definitely not buy 1 _____ 2 _____ 3 _____ 4 _____ 5 _____

B. Open-End Questions		
Completely unstructured	A question that respondents can answer in an almost unlimited number of ways	What is your opinion of American Airlines?
Word association	Words are presented, one at a time, and respondents mention the first word that comes to mind.	What is the first word that comes to your mind when you hear the following? Airline_____ American_____ Travel_____
Sentence completion	An incomplete sentence is presented and respondents complete the sentence.	When I choose an airline, the most important consideration in my decision is _____.
Story completion	An incomplete story is presented, and respondents are asked to complete it.	"I flew American a few days ago. I noticed that the exterior and interior of the plane had very bright colors. This aroused in me the following thoughts and feelings . . ." Now complete the story.
Picture	A picture of two characters is presented, with one making a statement. Respondents are asked to identify with the other and fill in the empty balloon.	
Thematic Apperception Test (TAT)	A picture is presented and respondents are asked to make up a story about what they think is happening or may happen in the picture.	



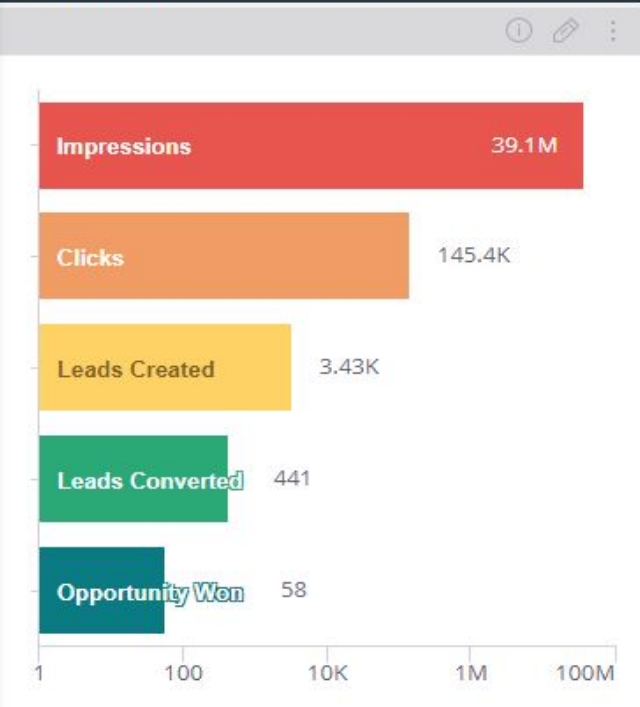
Measuring Marketing Productivity

Marketing metrics is the set of measures that helps them quantify, compare, and interpret their marketing performance.

Marketing-mix modeling Analyzes data from a variety of sources, such as retailer scanner data, company shipment data, pricing, media, and promotion spending data, to understand more precisely the effects of specific marketing activities

Marketing dashboards A concise set of interconnected performance drivers to be viewed in common throughout the organization

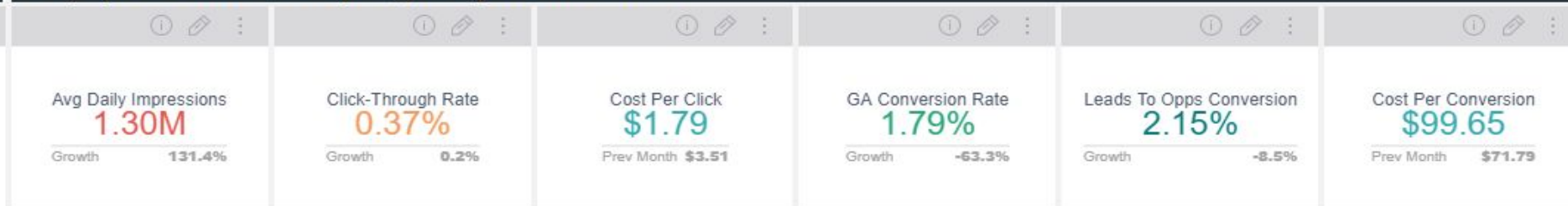
Funnel



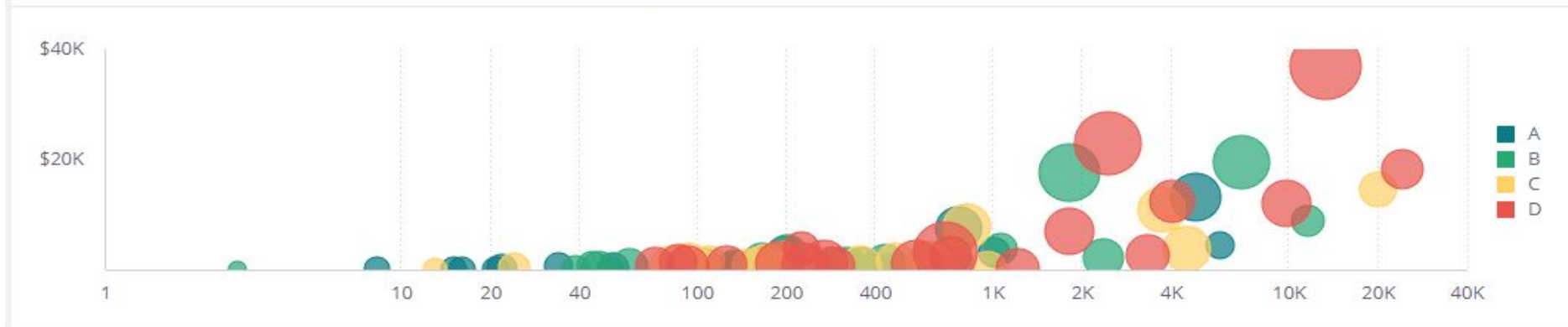
Top Campaigns



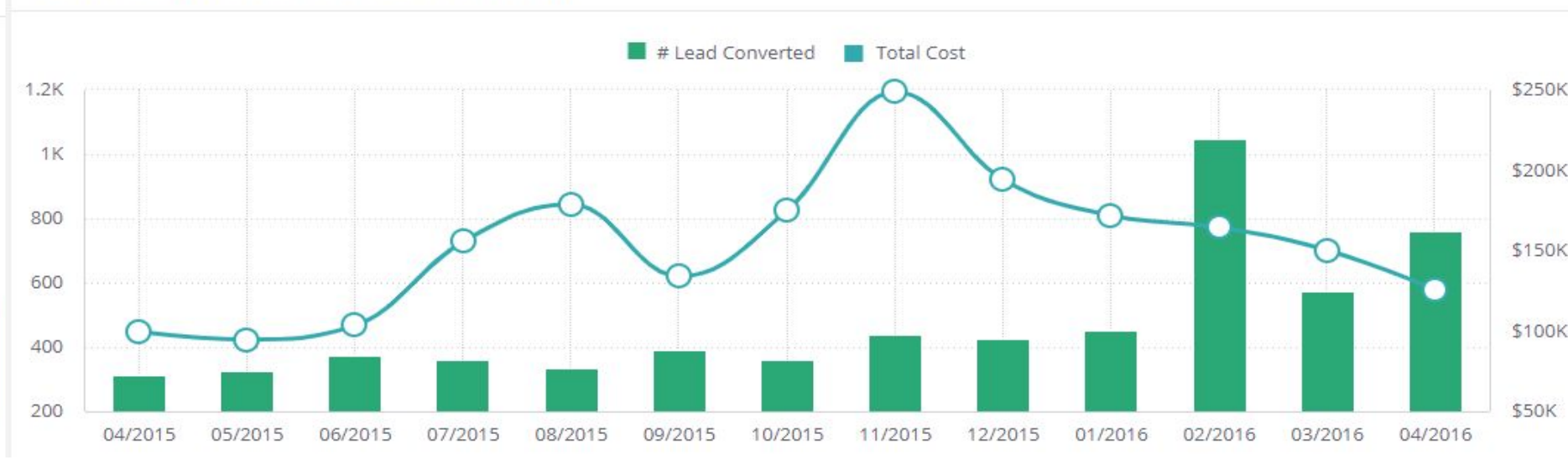
Pay-per-click Campaign Optimization



Cost to Interaction Optimization - By Lead Scoring (Highest: A | Lowest: D)



Leads Converted Vs. Cost Trend

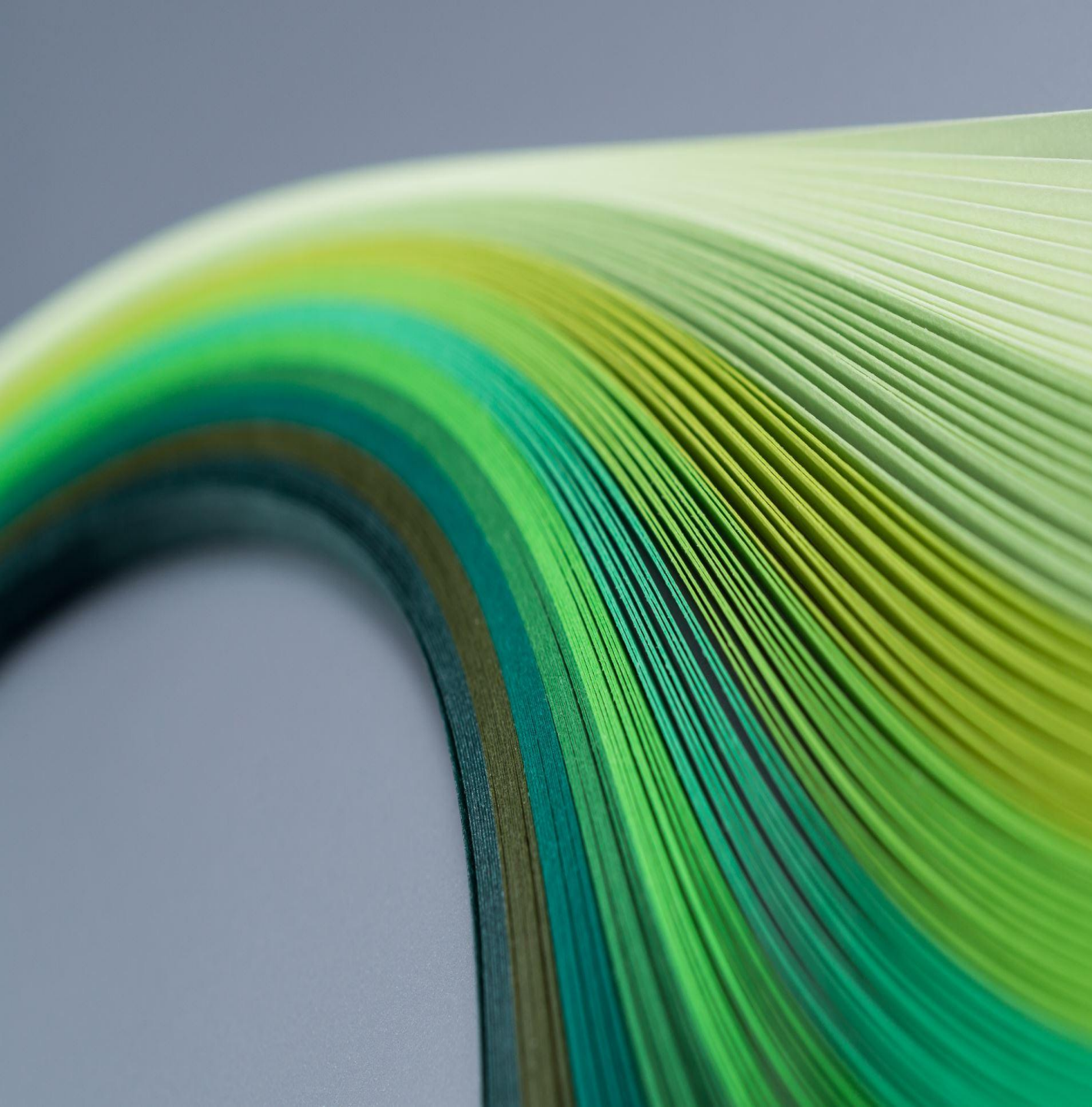


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Price Research

It is the process of collecting data on the prices of goods and services to help businesses decide what prices to charge, how much to produce, and what marketing strategies to use. Price research is used in a variety of different industries, including manufacturing, retail, and service industries.

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Media Research

Media Research is the study of the effects of the different mass media on social, psychological, and physical aspects. Research segments the people based on what television programs they watch, the radio they listen to, media they access, and the magazines they read.

Parameters in Media Research

- ✓ The nature of the medium being used
- ✓ The working of the medium
- ✓ Technologies involved in it
- ✓ Differences and similarities between it and other media vehicles
- ✓ Functions and services provided by it
- ✓ The cost associated and access to a new medium
- ✓ Effectiveness and how it can be improved

The effectiveness in terms of time spent, interactions, etc. are measured. It also includes the costs associated with each channel and the ROI provided.



How to gather data?



There are several different ways to collect price data



One way is to contact businesses directly and ask them for information on their prices.



Another way is to use public databases, such as the Bureau of Labor Statistics Consumer Price index(CPI).



The most common types of price research include market research, cost analysis, and competitor analysis

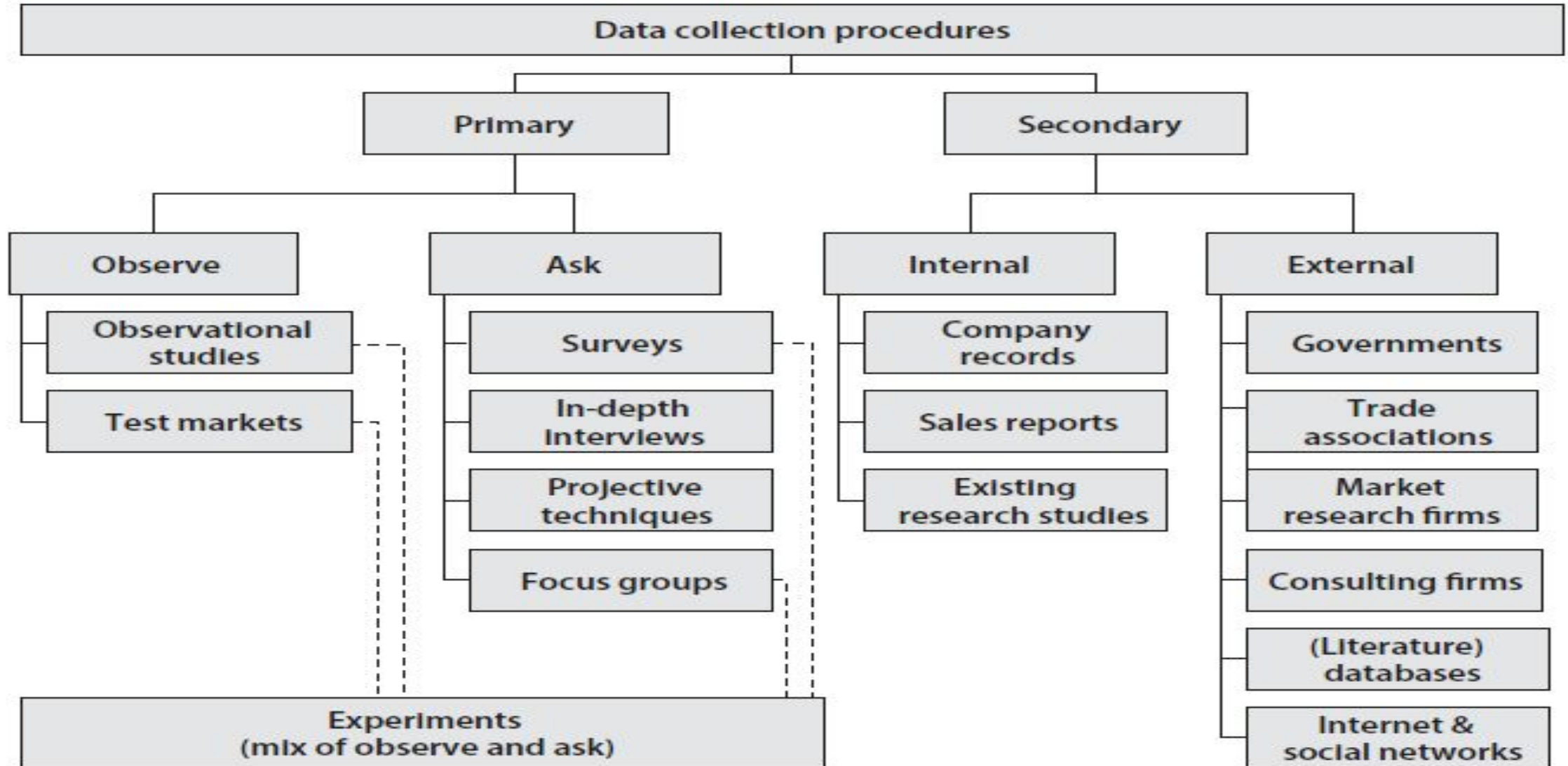
Data Collection

Data Sources – Marketing research data is classified under two broad categories– Primary data and Secondary data.

Primary data is the information that is collected directly using in-depth interviews, surveys, focus groups, and social media monitoring.

Secondary data is existing information in the public domain which has been collected by others. Secondary data can be collected from government-published statistics, publications of industry associations, trade and commercial publications, company websites, market research reports, or journal articles.

Data collection methods



Sampling Techniques

Sampling is one of the important components of research design. The process to be considered is

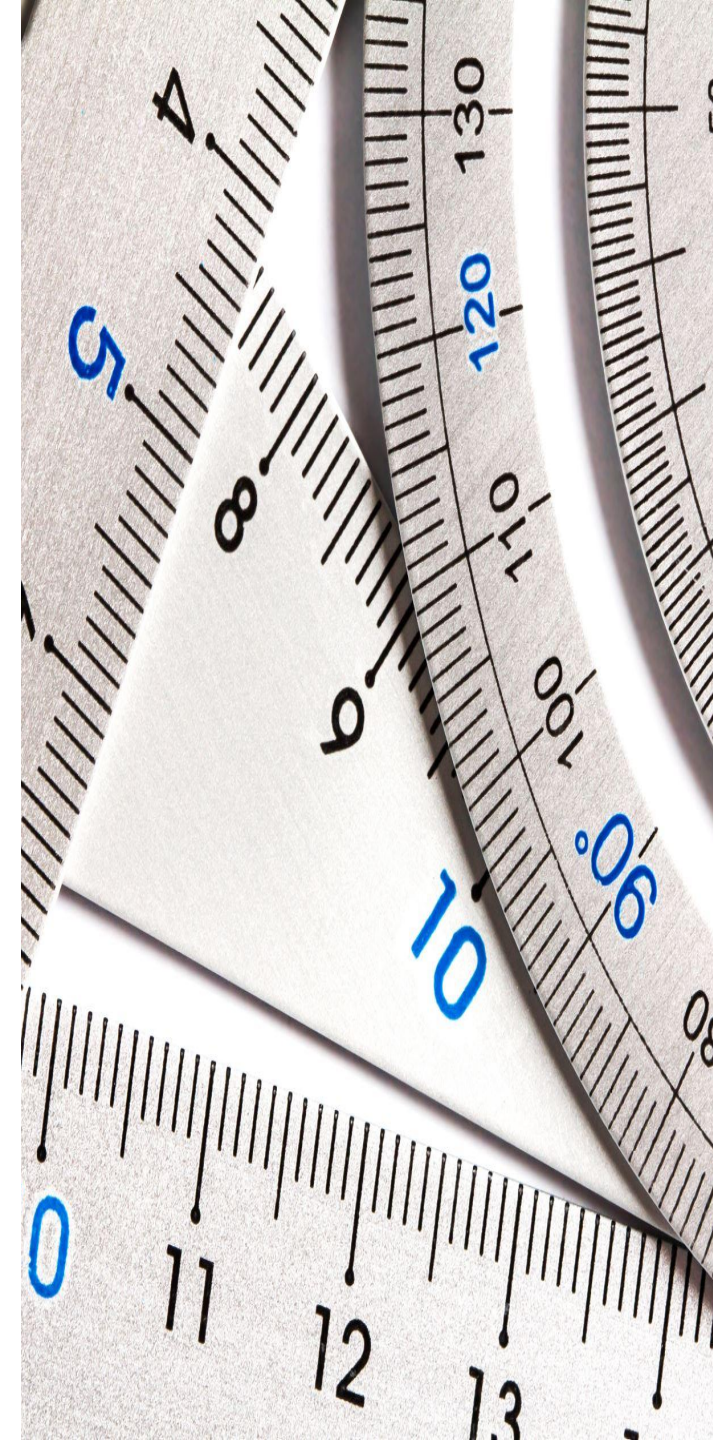
- i) Target Population – This corresponds to the complete group whose Marketing Research relevant characteristics are to be determined through the sampling.
- ii) Sampling Frame – This is the representation of the elements of the target population.
- iii) Sampling Technique – The sampling techniques or methods are broadly classified as probability sampling methods or nonprobability sampling methods. Probability sampling methods utilize random sampling techniques to create sample. On the other hand, non-probability sampling methods do not give all the units equal chance of being selected as part of the sample

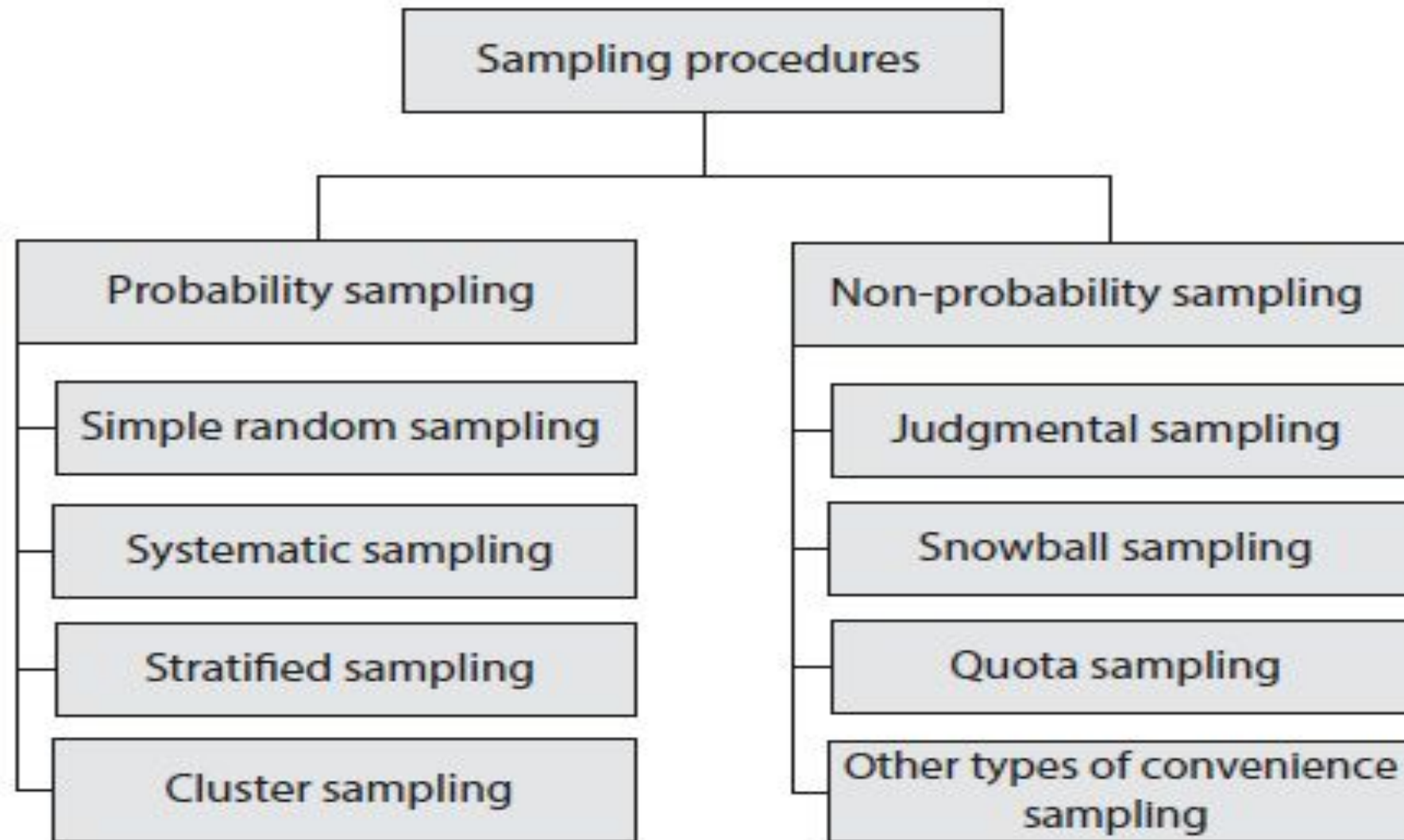


Iv) Sample Size – This corresponds to the sub-population to be studied to make the inference of the reference population. It is an important feature of the sampling process as the sample size should be adequate enough to represent the population under consideration.

Measurement scaling -There are four levels of measurement :

Nominal scale,
ordinal scale,
interval scale, and
ratio scale.





Probability or Random Sampling Methods:

- **(1) Simple random sampling** – This method simply involves the task selecting sampling units randomly out of the sampling frame. A researcher may use the following methods for selecting random samples – Lottery Method, Random Numbers, software etc.
- **(2) Stratified sampling** – In this method a heterogeneous population is divided into different small sub-units, which are called stratas. These stratas are homogenous among themselves with respect to a certain factor or characteristic. Items or sampling units are randomly selected from these stratas that together make up the sample.
- **3) Systematic sampling** – In this type of sampling the first unit is selected randomly and then every K^{th} item on the source list is selected, which becomes the part of the sample. The value of K is determined by :
 - $K = \text{Total no. of units in population} / \text{No. of units in sample}$
 - The essence of this method is selection of random items from the source list at a specified interval from the selected unit, hence forming a system for selecting items. The Items may be arranged numerically, alphabetically or in an increasing or decreasing order and then a formula is applied to it.
- **(4) Cluster sampling** – This method is used where the size of population is very large. In this method a homogeneous population is divided into smaller heterogeneous groups and then samples are drawn out at random from these heterogeneous groups. These heterogeneous groups are called clusters. All items belonging to the selected heterogeneous groups become the part of the sample.
- **(5) Area Sampling** – If the clusters are divided on geographical basis, it is termed as area sampling.
- **(6) Multi-stage sampling** – In multistage sampling, sampling is performed at more than 1 step or stage. At first stage units are selected by some random sampling method usually SRSWOR or Systematic sampling and at the second stage again some units are selected out of the previously selected units through some suitable method. It can be understood as an expansion of the cluster sampling method where instead of selecting the entire heterogeneous group, items are drawn randomly from each heterogeneous group to form a sample.

- **Non-Probability or Non-Random Sampling Methods**

- **(1) Judgement sampling** – In this method, the sampling units are chosen by the researcher on the basis of his or her own judgement. The research simply selects the sample which in his opinion will be best for the study.
- **(2) Quota sampling** – In this method of sampling, quotas in form of reservation or percentage are established for different classes of population on the basis of age, gender, nationality etc. A sample is then drawn out on the basis of these quotas.
- **(3) Panel sampling** – In this method regular surveys are taken by a researcher from a panel of experts of a particular domain through [questionnaires or schedules](#). The panelists may or may not know about other during the research process.
- **(4) Convenience sampling** – In convenience sampling, a researcher simply selects the sample and sampling units that are easily available and accessible. No extra efforts are taken by the researcher as he simply chooses the samples on the basis of convenience.
- **(5) Snowball sampling** – This method is used in cases where the population to be studied is rare, therefore it is difficult to find good representative sampling units. In this method the researcher initially selects a sampling unit (a doctor, a musician, a cancer patient depending upon the study) based on his judgement and then starts taking further samples on the basis of directions/advice/referral provided by the first sampling unit.

Differences between measurements, true zero exists

Ratio Data

Quantitative Data

Differences between measurements but no true zero

Interval Data

Ordered Categories (rankings, order, or scaling)

Ordinal Data

Qualitative Data

Categories (no ordering or direction)

Nominal Data



Scale	Characteristics	Uses in marketing	Possible statistics*
Nominal	Identity, single definition of numbers	Brands, Gender, Race, Colors, Types of stores, Regions, Use/non-use, Like/dislike, and any variable to which numbers can be associated for identification	Frequency distribution, Mode, Percentages, Binomial test, Chi-square test, McNemar, Cochran's Q
Ordinal	Numbers order	Attitudes, Preferences, Opinions, Social classes, Occupations	All the ones from nominal scale plus: Median, Quartiles, Deciles, Percentile, Mann-Whitney Test, U-Test, Kruskal Wallis, Correlation of Spearman or Kendall
Interval	Comparison of intervals	Attitudes, Opinions, Awareness, Preferences, Index Numbers	All the ones from nominal and ordinal scales plus: Average, Interval, Total amplitude, Average amplitude, Average deviation, Standard deviation, Z-Test, t-Test, F-Test, Analysis of variance, Pearson's correlation
Ratio	Comparison of absolute measures, Comparison and proportions	Age, Price, Number of consumers, Sales volume, Income, Inheritance	All the ones from nominal, ordinal and interval scales plus: Geometric average, Harmonic average and Coefficient of variation

* Appropriate statistics to a scale are cumulative the higher the scale level is, all are allowed in the ratio scale.

Data analysis methods

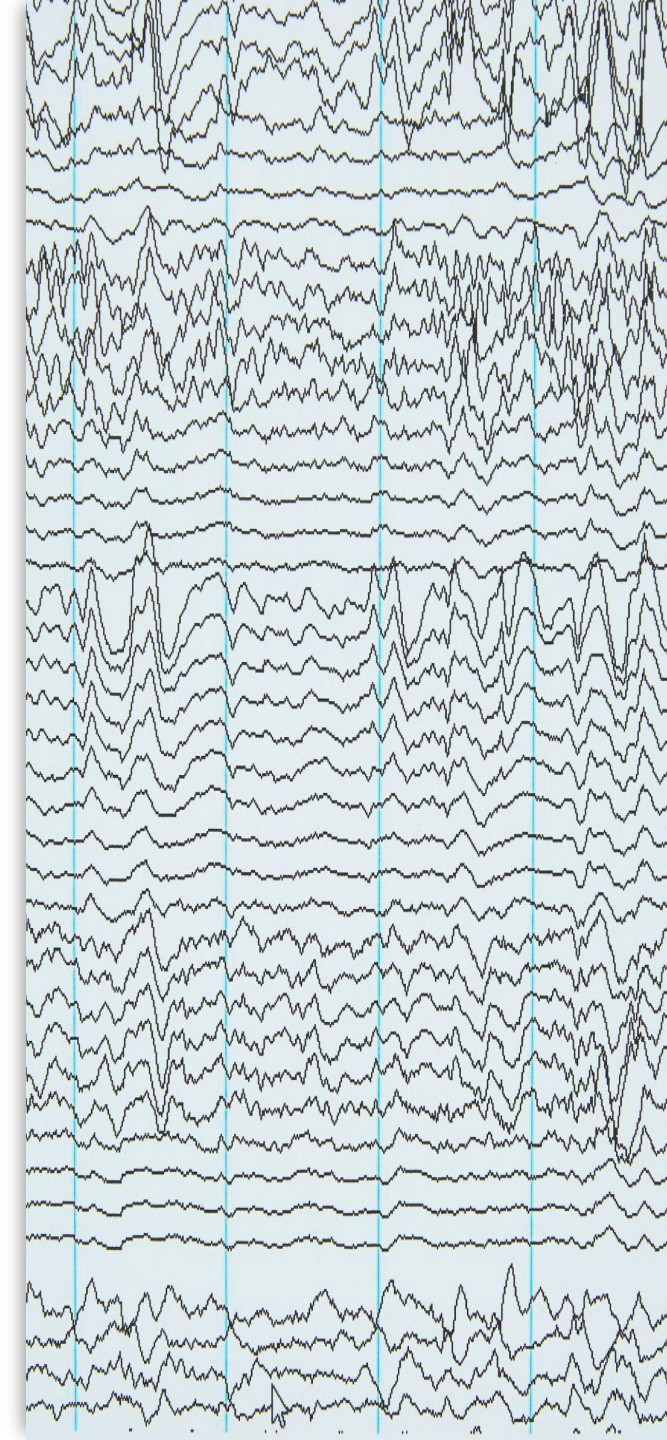
Data analysis is the process of converting the raw data into meaningful insights using relevant methods and techniques. The data analysis methods selected are based on the research approach i.e. quantitative vs. qualitative. The four types of data analysis that is typically useful are :

- ❖ descriptive analysis (What is happening?),
 - ❖ diagnostic analysis (Why is it happening?),
 - ❖ predictive analysis (What is likely to happen?); and
 - ❖ prescriptive analysis (What do I need to do
-
- ❖ In quantitative research designs, three types of analysis are generally done – Marketing Research
 - ❖ uni-variate analysis (analysis based on single variable);
 - ❖ bi-variate analysis (using two variables simultaneously); and
 - ❖ multi-variate analysis (examination of more than two variables).

The various descriptive statistics that are calculated are – range, average, median, mode, variance, standard deviation, so on.

Various methods include t-test, Z-test, One way ANOVA, Chi-Square test. In comparison to univariate analysis, bi-variate analysis constitutes comparison by analyzing two variables simultaneously.

The basic objective is to determine relationships between two variables. Examples of such methods are correlation and simple regression analysis. A simultaneous study of more than two variables is called as multivariate analysis.



Validity and reliability

- The systematic and random error concepts are important because they relate to a measure's validity and reliability. Validity refers to whether we are measuring what we want to measure and, therefore, to a situation where the systematic error ES is small.
- Reliability is the degree to which what we measure is free from random error and therefore relates to a situation where the ER is zero.

Understanding Hypothesis Testing

Hypothesis testing starts with the formulation of a null and alternative hypothesis. A null hypothesis (indicated as H_0) is a statement expecting no difference or effect. Conversely, an alternative hypothesis (indicated as H_1) is the hypothesis against which the null hypothesis is tested.



There are several tools available like multiple regression, factor analysis, cluster analysis, discriminant analysis, conjoint analysis, multi-dimensional scaling, structural equation modeling, and many other complicated techniques.

For data analysis purposes, the marketing researcher can use various software like SPSS, SAS, AMOS, SmartPLS, and R-software. Similarly, for qualitative research data analysis, the researcher can use software like NVivo, ATLAS, Ti.



Multivariate analysis, Factor analysis, Cluster analysis,



Multivariate analysis finds patterns and correlations between multiple factors by analyzing two or more variables at once.



Factor Analysis is used to explain correlation in a set of data and relate variables to each other.



Cluster Analysis is used to group similar objects together based on their similarity or distance

Discriminant analysis & Conjoint analysis

Discriminant analysis (DA) is a multivariate technique which is utilized to divide two or more groups of observations (individuals) premised on variables measured on each experimental unit (sample) and to discover the impact of each parameter in dividing the groups.

Conjoint analysis is one of the most effective models for extracting consumer preferences during the purchasing process. This data is then turned into a quantitative measurement using statistical analysis. It evaluates products or services in a way no other method can

